Harder Geometrical Problems — problem 12.3.2.81

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In a tetrahedron $A_1A_2A_3A_4$, the face opposite vertex A_i has area S_i. Choose a point P inside the tetrahedron with distances $x_1, \ldots, x_4$ from the faces $S_1, \ldots, S_4$, respectively, such that the sum $\sum S_i/x_i$ is minimal.